VM and Database as a Service (DBaaS)

Cloud Computing - Winter 2026 - F18L3W068

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PART 1:
Problem & Environment

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Project & Problem Description

Overview:

- Proof-of-Work (PoW) Job Processing Service deployed across two cloud providers.
- Application layer runs on Google Cloud Compute Engine.
- Persistent Database via Azure PostgreSQL.

Purpose of the system is to process computationally expensive PoW jobs submitted by clients through an HTTP API.

Project & Problem Description

Purpose of the system is to **process computationally expensive PoW jobs** submitted by clients through an HTTP API.

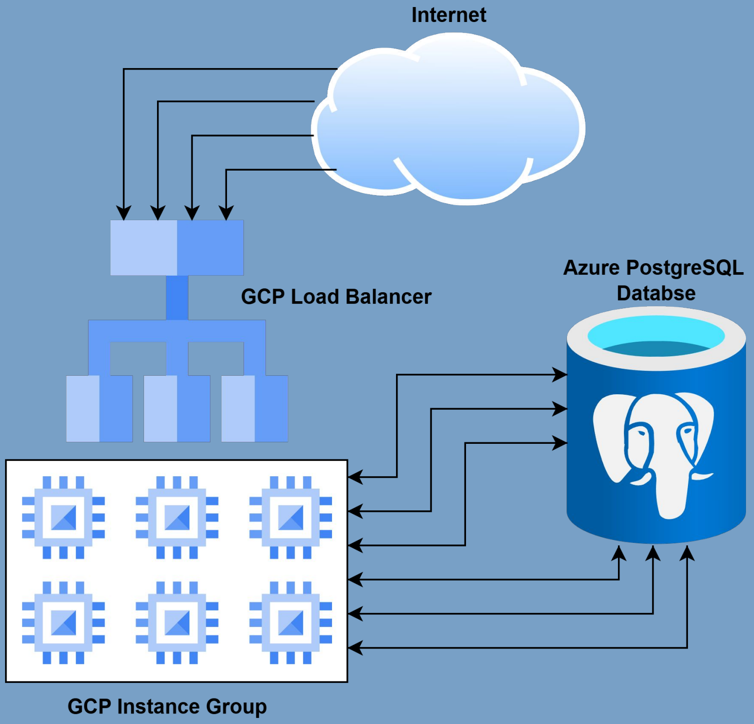
- A job consists of input data and a difficulty value.
- System must find a numeric nonce such that the SHA-256 hash of **`input_data + : + nonce`** starts with a required number of **leading zero** characters.
- The required number of zeroes is equal to the difficulty.

For example, if the difficulty is 6, the resulting hash must begin with `000000`...

Project Layout

Regular workflow:

1) Internet client sends a PoW.
2) Load Balancer forwards the request to a healthy VM instance.
3) Instance inserts a new job into the DB.
4) Worker processes poll the DB to claim queued jobs.
5) Worker performs the PoW computation.
6) Result, nonce, and execution info is written back to APDB.
7) The client retrieves the job status or final result via the API.



Infrastructure Characteristics

Infrastructure characteristics	**GCP Compute Engine VM template**	**Azure PostgreSQL Flexible Server**
Service role	Application and PoW workers	Managed database and job queue
Instance type	e2-standard-2	Standard_B1ms
vCPU / vCore	2 vCPUs per VM	1 vCore
Memory	8 GiB per VM	2 GiB
Storage	Boot persistent disk for container/runtime	32 GiB database storage
Scaling used in benchmark	1–6 VMs through MIG	Single managed DB instance

The GCP values are per VM template.

During the benchmark, the Managed Instance Group scaled up to six VMs.

The maximum compute capacity was **12 vCPUs** and 48 GiB memory.

PART 2:
Experiments & Results

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Workload Generator

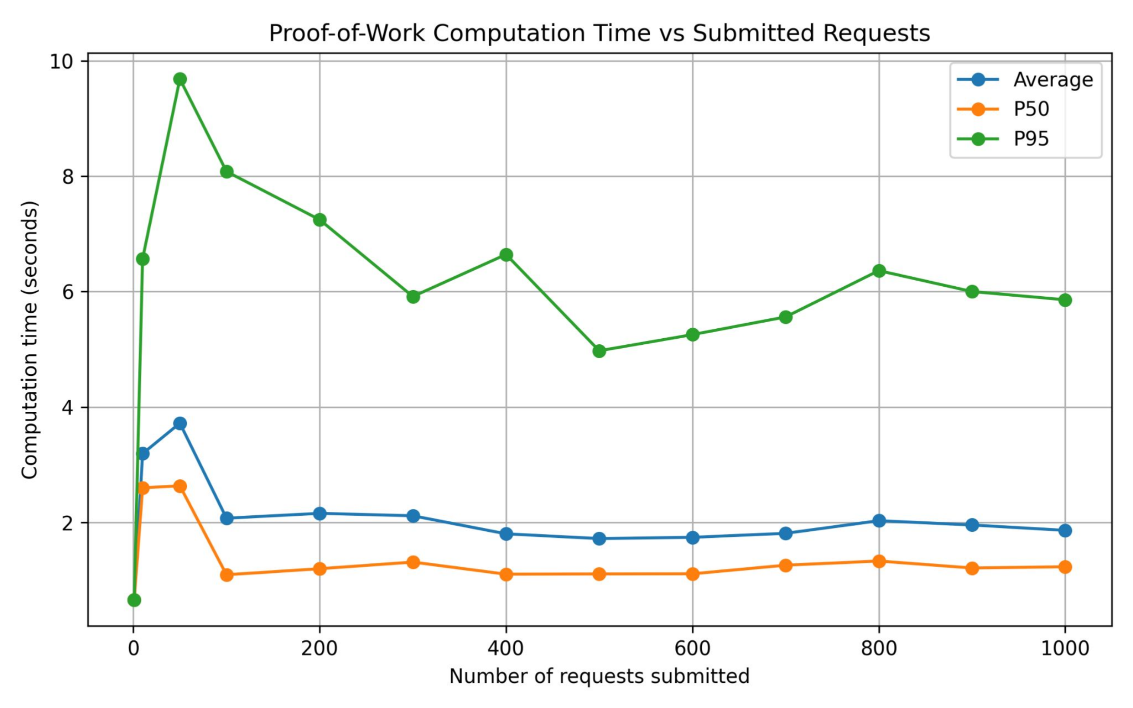
A custom duration-based workload generator was used to evaluate the Proof-of-Work service under increasing request rates.

- Tested request rates: 1, 10, 50, 100, 200, 300, 400, 500, 600, 700, 800, 900, 1000 (rps).
- Duration per rate was set to 1 second.
- Each submitted job used PoW difficulty 5.
- The difficulty is configured so that each job takes ≈2 seconds of **computation time**, not accounting for **queue wait time**.

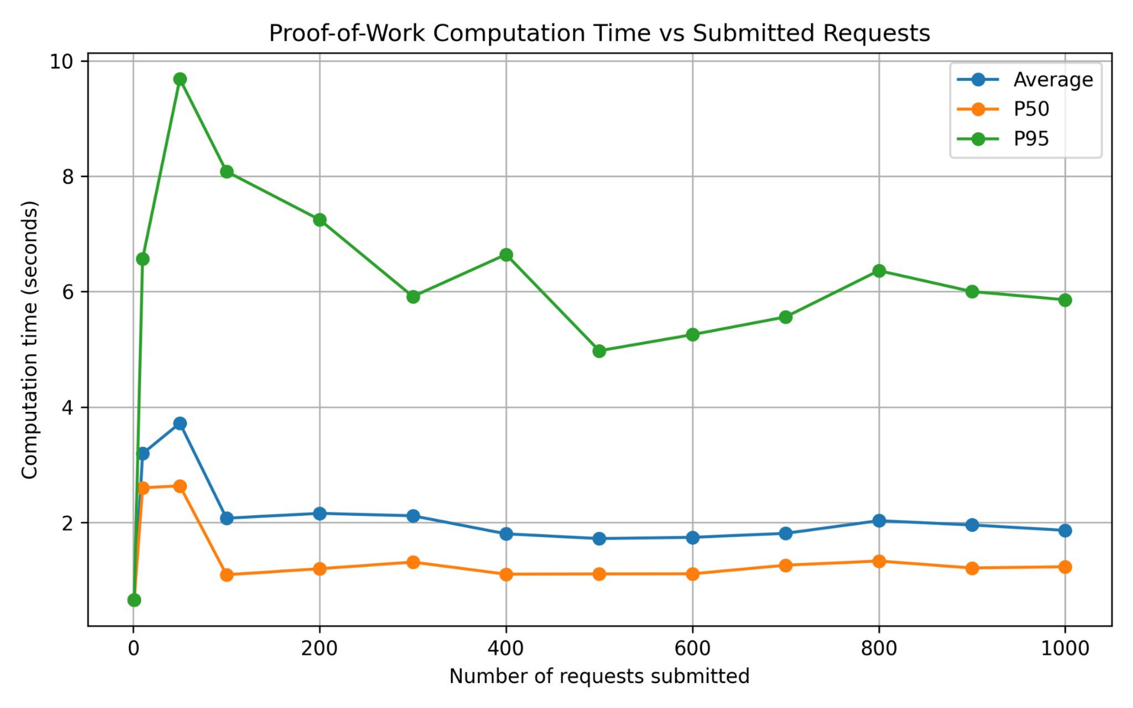
⚠️*A duration of 1 was chosen, due to the hard limitation of 12 allowed vCPUs* ⚠️

Durations larger than 1 **severely skew the results** making them much harder to interpret, while making the source of **a major bottleneck** more prevalent.

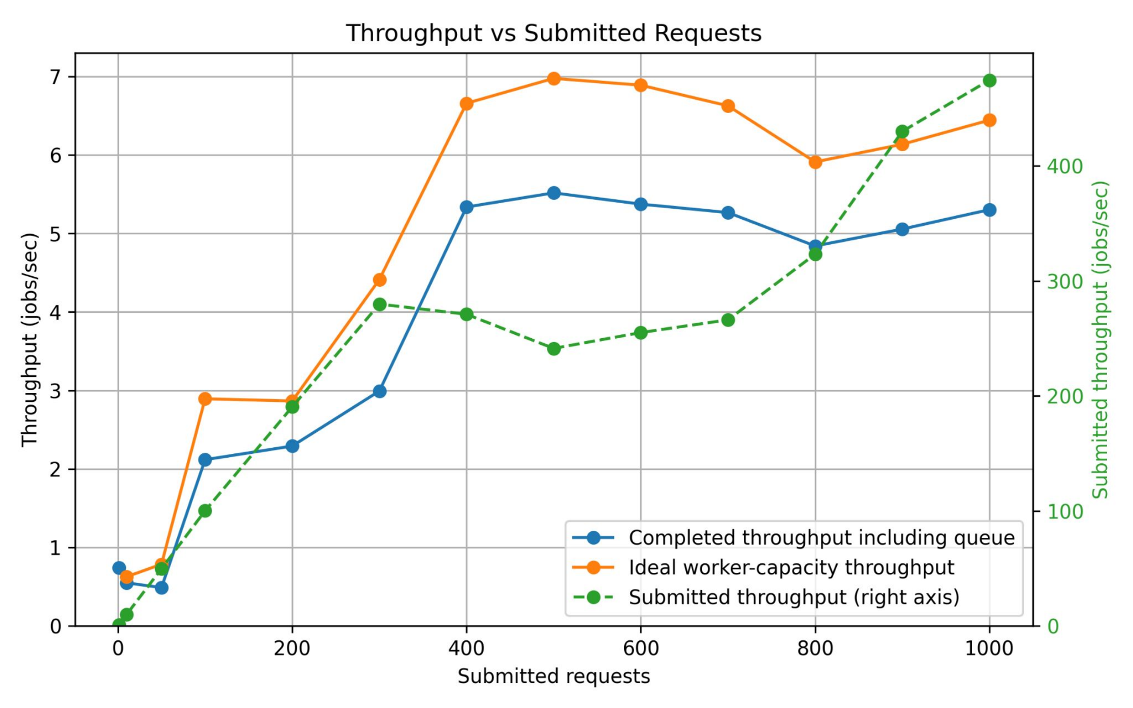
Execution Time



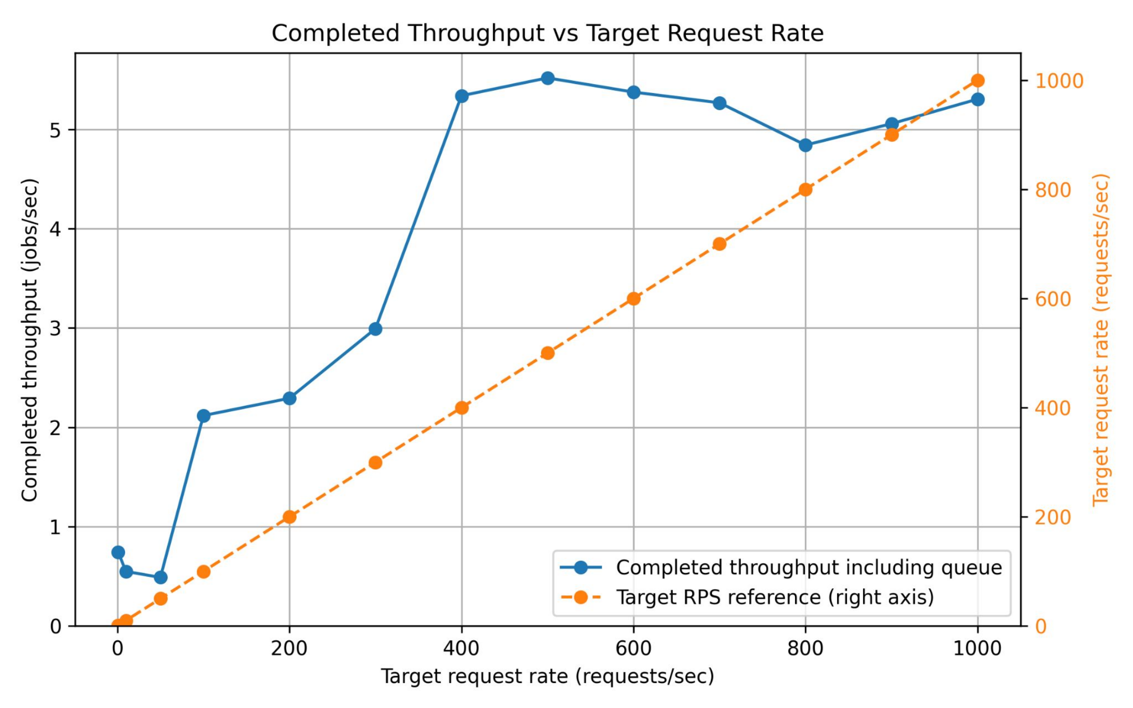
Execution Time



Throughput

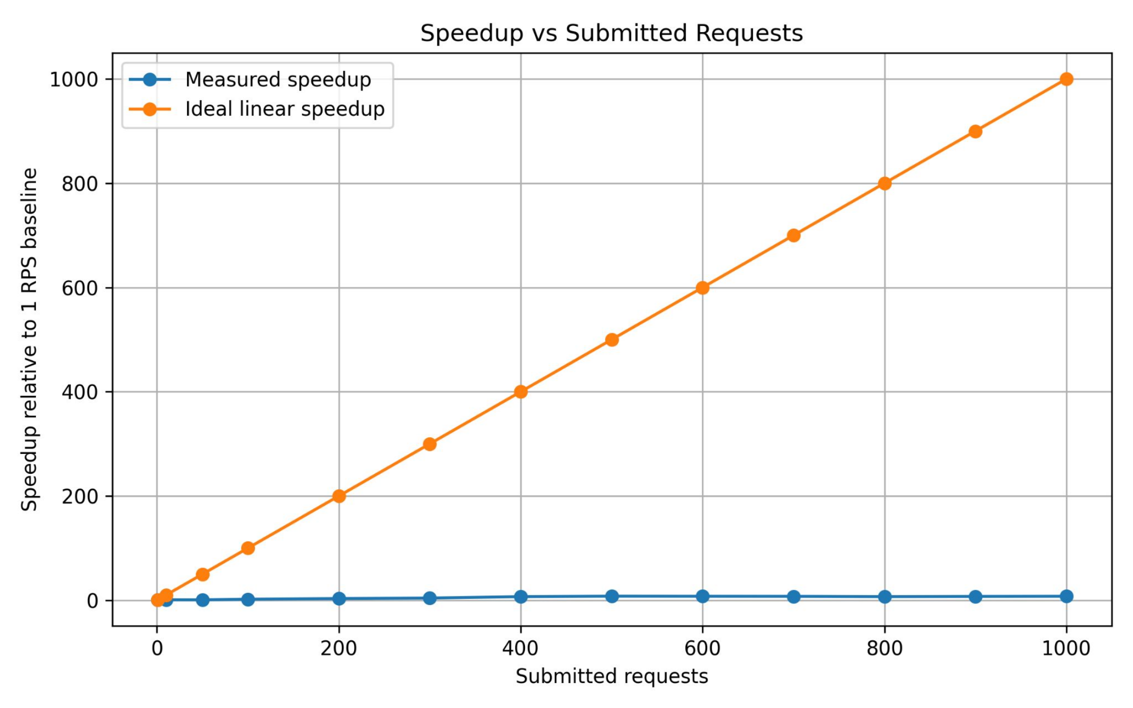

Throughput vs Submitted Requests
Throughput (jobs/sec)
Submitted throughput (jobs/sec)
Submitted requests
0
1
2
3
4
5
6
7
0
200
400
600
800
1000
100
300
Completed throughput including queue
Ideal worker-capacity throughput
Submitted throughput (right axis)

Throughput



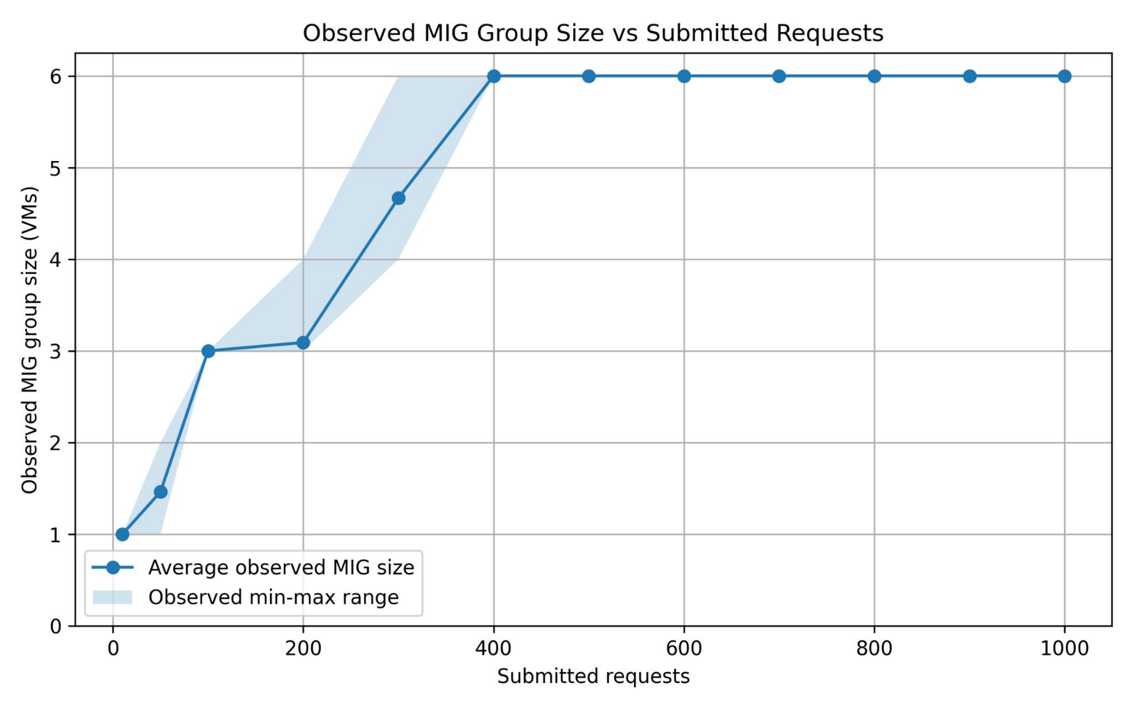
Completed Throughput vs Target Request Rate
Completed throughput (jobs/sec)
0
1
2
3
4
5
Target request rate (requests/sec)
0
200
400
600
800
1000
Target request rate (requests/sec)
0
200
400
600
800
1000
Completed throughput including queue
Target RPS reference (right axis)

Speedup

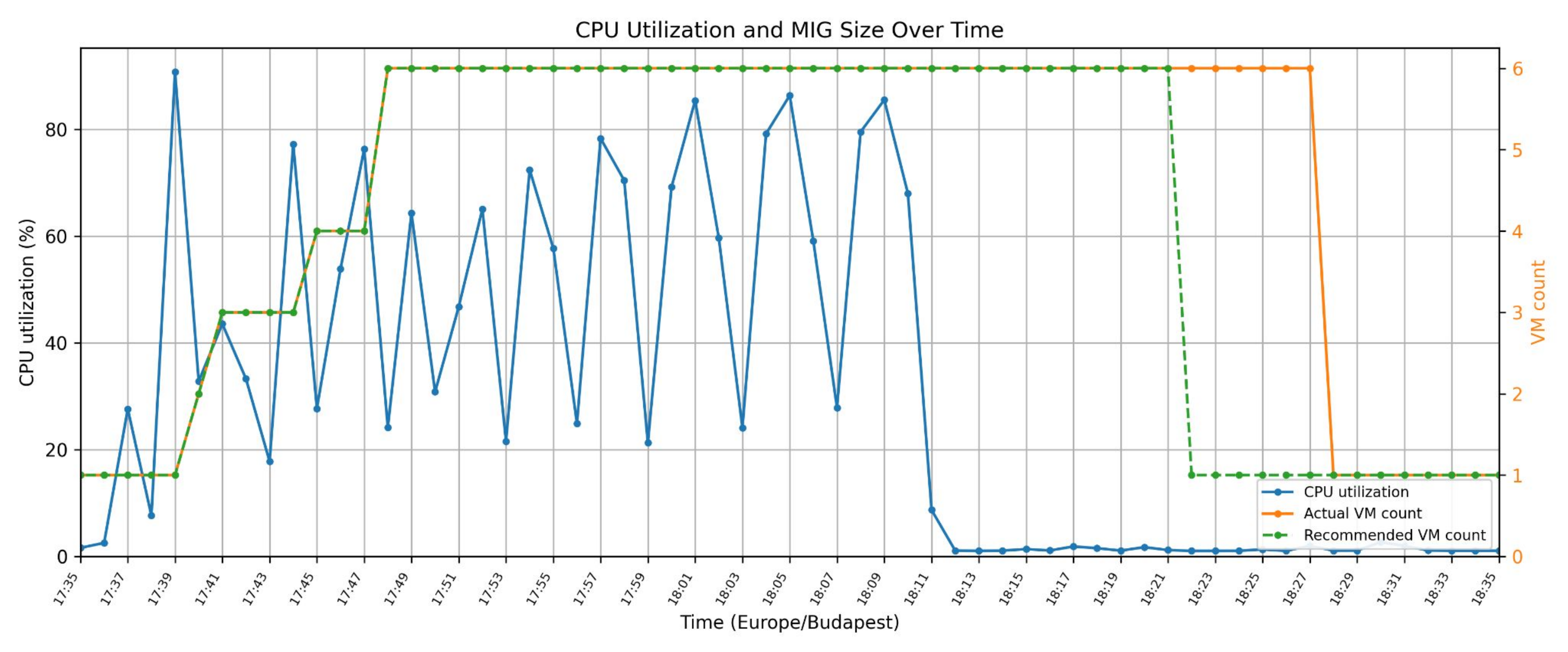


Speedup vs Submitted Requests
Measured speedup
Ideal linear speedup
Speedup relative to 1 RPS baseline
Submitted requests
0
200
400
600
800
1000

Scalability



Elasticity & Scalability



Cost Analysis

Component Cost	(USD)
Azure PostgreSQL Flexible Server	0.028
Google Compute Engine VMs	0.306
Cloud NAT gateway	0.006
Cloud NAT external IP address	0.005
External LB forwarding rule	0.023
Artifact Registry	0.000
Total	0.368

Final Evaluation

Metric	Very short description	Final evaluation
Execution time	End-to-end request completion time, including queueing and PoW processing.	POOR
PoW computation time	Actual worker processing time after a job is claimed.	GOOD
Throughput	Completed proof-of-work jobs per second.	GOOD
Speedup	Throughput improvement compared with the low-load baseline.	POOR
Scalability	Ability to increase capacity as request load grows.	GOOD
Elasticity	Ability of the MIG to scale out and scale back down automatically.	GOOD
Cost	Estimated infrastructure cost for the benchmark run.	EXCELLENT

Thank You for Your Attention!

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Any Questions?